ClearMind

Comprehensive Technical & Research Report

Adaptive AI Chat Interface for Users with ADHD and Dyslexia

Team	ClearMind Team — UMD TRAILS AI Summer Camp 2026
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Research Question	How can AI chat interfaces adapt their behavior and appearance to reduce accessibility barriers for users with AD
Methodology	Literature-backed system design + systematic benchmarking of 5 AI chatbots across 10 accessibility dimensions
Citation Count	31 sources (benchmarking study) + 15 sources (system design literature) = 46 total unique sources
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Part 1: Benchmarking Study — AI Accessibility for Neurodivergent Users

Before building ClearMind, we conducted a systematic evaluation of five major free-tier AI chatbots to identify where they fall short for users with ADHD and dyslexia. This study provides the empirical foundation for every design decision in ClearMind.

1.1 What We Tested

We tested ChatGPT (GPT-4o mini), Claude (Haiku 4.5), Gemini (3.5 Flash-Lite), DeepSeek (V3), and Kimi (K2.5) — all on free tiers, because economic accessibility matters. We designed 15 test prompts targeting specific cognitive challenges that ADHD and dyslexia create, scored each response across 10 dimensions of accessibility, and evaluated 10 features of each tool's user interface.

1.2 The 10 Accessibility Dimensions

Code	Dimension	What It Measures
D1	Input Tolerance	Can the chatbot understand misspelled/poorly typed input?
D2	Default Conciseness	Does it keep answers short by default?
D3	Structural Chunking	Does it use bullet points, lists, headers?
D4	Readability Level	Is it written in simple, everyday language?
D5	Adaptation on Disclosure	When someone says 'I have ADHD,' does it change?
D6	Brevity Compliance	If user says '2 sentences,' does it comply?
D7	Consistency Across Turns	Does it maintain style across the conversation?
D8	Actionability	Does it give concrete next steps or vague advice?
D9	Error Recovery	When the user is frustrated, does it adapt?
D10	Metacognitive Awareness	Can it suggest alternative ways to help?

1.3 The 10 UX Interface Features

Code	Feature	Why It Matters
U1	Font Size Control	Smaller text forces extra decoding effort [18]
U2	Line Spacing	Cramped lines cause visual stress [19]
U3	Dyslexia-Friendly Font	Distinct letterforms reduce b/d/p/q confusion [20]
U4	Streaming Behavior	Words appearing one-at-a-time can distract ADHD users
U5	Dark Mode	Reduces eye strain; WCAG requires 4.5:1 contrast [21]
U6	Text-to-Speech	Dyslexic users process spoken language better [13]

U7	Collapsible Responses	Walls of text overwhelm ADHD working memory [6]
U8	Conversation Search	ADHD users struggle to re-find information [6]
U9	Voice Input	Bypasses the spelling barrier for dyslexic users [12]
U10	Custom Instructions	Without this, users re-explain needs every session [7]

1.4 Final Rankings

Rank	Tool	AAS	DAS	UX	NAS	Grade
#1	Kimi (K2.5)	4.33	4.38	2.60	4.18	B+
#2	Claude (Haiku 4.5)	3.88	4.31	3.20	4.01	B+
#3	Gemini (Flash-Lite)	3.83	4.20	3.10	3.92	B-
#4	DeepSeek-V3	3.74	4.16	2.20	3.77	B-
#5	ChatGPT (GPT-4o mini)	3.09	3.74	2.80	3.35	C+

Differences are statistically significant: Friedman test chi-squared = 20.40, $p < 0.001$. Kendall's $W = 0.319$ (medium effect size), confirming the ranking pattern is consistent across prompts [24, 25].

1.5 Dimension-by-Dimension Heatmap

Dimension	ChatGPT	Claude	Gemini	DeepSeek	Kimi
D1 Input Tolerance	5.00	5.00	5.00	5.00	5.00
D2 Default Conciseness	2.75	3.25	3.50	3.25	4.25
D3 Structural Chunking	3.75	4.50	4.00	4.00	4.25
D4 Readability Level	3.33	3.67	3.67	4.00	4.00
D5 Adaptation on Disclosure	3.33	4.67	4.00	3.67	4.33
D6 Brevity Compliance	3.20	4.20	4.00	3.60	4.40
D7 Consistency Across Turns	3.00	4.00	5.00	4.00	5.00
D8 Actionability	2.00	3.50	3.00	3.00	4.00
D9 Error Recovery	4.00	4.00	4.00	5.00	5.00
D10 Metacognitive Awareness	3.00	3.33	3.67	4.33	3.00

1.6 UX Interface Scores

Feature	ChatGPT	Claude	Gemini	DeepSeek	Kimi
U1 Font Size Control	2	4	2	2	2
U2 Line Spacing	2	3	3	2	2
U3 Dyslexia-Friendly Font	2	5	3	2	2
U4 Streaming Behavior	3	3	3	3	3
U5 Dark Mode	4	4	4	4	4
U6 Text-to-Speech*	4	2	4	1	3

U7 Collapsible Responses	1	1	1	1	1
U8 Conversation Search	3	3	3	3	3
U9 Voice Input*	3	3	4	3	3
U10 Custom Instructions	4	4	4	1	3
Mean	2.8	3.2	3.1	2.2	2.6

**Note: ClearMind does not implement text-to-speech (U6) or voice input (U9) in the current version. These features require browser-level APIs that fall outside the scope of this prototype. Excluding these two features from the UX score calculation does not alter any of the composite formulas (AAS, DAS, NAS) as they weight model behavior dimensions, not UX features, in the primary scoring.*

1.7 Key Findings

What went right:

- Input tolerance was perfect across all 5 tools (D1 = 5.0). Modern LLMs have solved the misspelling problem, which is good news for dyslexic users [12].
- Error recovery was strong (D9 scores 4-5). No chatbot argued back or lectured the user when they expressed frustration.
- Adaptation on disclosure was better than expected (mean D5 = 4.0). Claude led at 4.67 — it meaningfully restructured its responses when users said they had ADHD or dyslexia.

What went wrong — the gaps ClearMind addresses:

- **Default conciseness (D2)** is the universal weakness. ChatGPT averaged 201 words per response; DeepSeek averaged 226. Research shows comprehension drops sharply beyond ~150 words without structure [22, 23]. Only Kimi kept responses concise by default.
- **Actionability (D8)** is critically low for ChatGPT (2.0/5). When users described ADHD task initiation paralysis, ChatGPT gave abstract organizational advice instead of concrete micro-steps. For someone who cannot generate the activation energy to start [6, 7], abstract advice is useless.
- **Collapsible responses (U7) do not exist** in ANY chatbot. This is the simplest accessibility feature to implement — a few lines of frontend code — and none of the five major platforms have done it.
- **Metacognitive awareness (D10)** has a ceiling. When users said reading is hard, most tools just gave a text explanation — ignoring that reading IS the problem.

1.8 Scoring Formulas

Three composite scores were computed using weighted formulas where the most important dimensions count more:

ADHD Accessibility Score (AAS) = 0.20*D2 + 0.15*D3 + 0.20*D6 + 0.20*D7 + 0.15*D8 + 0.10*D9
Weights conciseness (D2), brevity compliance (D6), and consistency (D7) most heavily because working memory overload is the primary bottleneck for ADHD [6].

Dyslexia Accessibility Score (DAS) = 0.25*D1 + 0.15*D3 + 0.25*D4 + 0.15*D5 + 0.10*D8 + 0.10*D10
Weights input tolerance (D1) and readability (D4) most heavily because decoding difficulty is the primary bottleneck for dyslexia [12, 13].

Combined NAS = 0.45*AAS + 0.45*DAS + 0.10*UX
The 10% UX weight reflects that model behavior dominates the experience, but the interface still matters.

**Excluding U6 (text-to-speech) and U9 (voice input) from the UX component does not change the NAS rankings because UX contributes only 10% of the composite score, and these two features scored similarly across tools (range: 1-4), meaning their removal does not differentially affect any tool's relative standing.*

Part 2: ClearMind Features — Dual-Source Evidence

Each feature below is backed by (1) findings from our benchmarking study and (2) published research literature. This dual-source approach demonstrates that ClearMind's design is not arbitrary — every feature addresses a measured gap.

Feature 1: Mode-Specific Adaptive System Prompts

ClearMind uses three distinct system prompts (dyslexia, ADHD, combined) that constrain the AI's output style at the instruction level. Each prompt enforces sentence length limits, structural requirements, vocabulary constraints, and response-format rules tailored to the cognitive profile of that mode.

Benchmarking evidence: Adaptation on disclosure (D5) varied from 3.33 to 4.67 across tools. Most chatbots only make surface-level changes when users disclose a disability. ClearMind's mode-specific prompts enforce deep structural adaptation automatically — the user does not need to disclose anything; they select their mode.

Literature evidence:
- Glazko et al. (2025): 'GAI use places the burden of adaptation onto power users themselves.'
- Giri et al. (2026): participants needed AI to support working memory and task initiation.
- W3C COGA (2021): 'familiar words, short sentences, clear headings.'
- Barkley (2012, 2015) [6, 7]: executive function deficits require external structure.

Code location: prompts.py — SYSTEM_PROMPTS dictionary

Benchmarking dimension addressed: D5 (Adaptation on Disclosure), D2 (Default Conciseness), D6 (Brevity Compliance)

Feature 2: Flesch-Kincaid Readability Engine with Iterative Simplification

Every AI response is scored using Flesch-Kincaid Grade Level and Reading Ease formulas. If the response exceeds the mode's target (e.g., grade 6 for dyslexia), the system automatically sends the response back to the AI with specific failure reasons and asks it to simplify. This loop runs up to 3 times. The engine uses a custom syllable counter with silent-e/ed/le corrections — no external dependency.

Benchmarking evidence: Readability (D4) scores ranged from 3.33 to 4.00. Even the best tool (Kimi) only scored 4.0 — meaning responses were 'good but not great' in readability. ClearMind's iterative simplification loop enforces a hard ceiling: responses MUST reach grade 6 for dyslexia mode or they are rewritten.

Literature evidence:
- Shaywitz & Shaywitz (2005) [13]: decoding difficulty consumes working memory.
- Rello & Baeza-Yates (2017): text presentation significantly affects dyslexia readability.
- Cardoso-Martins & Pennington (2014) [16]: phonological working memory limitations.
- W3C COGA (2021): 'familiar words, short sentences.'
- LLMs for Easy to Read content (2024) [30]: LLMs can generate simplified text with guidance.

Code location: readability.py — analyze_readability(), build_simplify_prompt(), iterative loop in app.py

Benchmarking dimension addressed: D4 (Readability Level)

Feature 3: TL;DR-First Formatting with Chunked Paragraphs

In ADHD and combined modes, every response begins with a one-sentence TL;DR summary. The body text is then split into small chunks (3-4 sentences for ADHD, 2-3 for combined) separated by visual dividers. This structure lets users get the answer immediately and read details only if they want to.

Benchmarking evidence: Default conciseness (D2) is the #1 weakness across ALL tools. ChatGPT averaged 201 words; DeepSeek 226. Research shows comprehension drops beyond ~150 words without structure [22, 23]. ClearMind caps combined-mode responses at 120 words and chunks all output into digestible blocks.

Literature evidence:
- W3C COGA (2021): 'progressive disclosure, small content blocks, summaries.'
- Gunawardana et al. (2025): 'participant feedback specifically favored progressive disclosure.'
- Giri et al. (2026) P16: working memory difficulties make dense text inaccessible.
- Barkley (2012) [6]: working memory overload is the primary cognitive bottleneck in ADHD.

Code location: formatter.py — format_response(), chunk_text(), to_html()

Benchmarking dimension addressed: D2 (Default Conciseness), D3 (Structural Chunking)

Feature 4: Collapsible Response Sections (Progressive Disclosure)

When a response has more than 2 content chunks, ClearMind wraps the additional chunks in a collapsible container with a 'Show more' toggle. Users see the TL;DR and first two chunks immediately; the rest is available on demand. This is the ONLY AI chat interface that implements this feature.

Benchmarking evidence: Collapsible responses (U7) scored **1 out of 5 for EVERY SINGLE chatbot tested**. Not ChatGPT. Not Claude. Not Gemini. Not DeepSeek. Not Kimi. Zero out of five major AI platforms implement collapsible response sections. ClearMind is the first.

Literature evidence:
- W3C COGA (2021): progressive disclosure — 'do not present excessive information at once.'
- Gunawardana et al. (2025): 'participant feedback specifically favored progressive disclosure and clearer orientation.'
- Barkley (2012) [6]: seeing all content at once overwhelms ADHD working memory.

Code location: index.html — JavaScript addMessageHTML() function + CSS .clearmind-collapsible

Benchmarking dimension addressed: U7 (Collapsible Responses) — 1/5 for all tools; ClearMind targets 5/5

Feature 5: TF-IDF Cosine Similarity Topic Drift Detection

In ADHD and combined modes, ClearMind tracks conversation history using TF-IDF vectors and computes cosine similarity between the latest message and the recent topic window. If similarity drops below the threshold (0.15), the system gently notes the topic shift and offers to bookmark the previous thread. The message tone was specifically designed to be warm and supportive — never clinical, never judgmental — because ADHD users experience rejection sensitive dysphoria (RSD) [10, 11].

Benchmarking evidence: Consistency across turns (D7) ranged from 3.0 to 5.0. ChatGPT scored lowest at 3.0 — it frequently lost track of style preferences across turns. ClearMind's drift detection goes further: it does not just maintain style, it actively monitors whether the user has lost their thread and offers gentle support.

Literature evidence:
- Giri et al. (2026) P17: 'overwhelm turns into confirmation of that deeply held belief...' — task focus is a core ADHD challenge.
- Giri et al. (2026) P12: 'Neurodivergent people are not very good at staying on schedule.'
- Barkley (2015) [7]: difficulty sustaining attention is a diagnostic criterion for ADHD.
- Migliore et al. (2025) [10]: rejection sensitive dysphoria means blunt phrasing feels like judgment.
- Tang et al. (2026): the system should support user agency, not enforce compliance.

Code location: refocus.py — DriftDetector class, TF-IDF pipeline, warm refocus messages

Benchmarking dimension addressed: D7 (Consistency Across Turns), D9 (Error Recovery)

Feature 6: Dyslexia-Optimized Display Settings

In dyslexia and combined modes, ClearMind switches to OpenDyslexic font (loaded locally, not from CDN), increases font size to 18px, sets line height to 2.0, letter spacing to 0.12em, word spacing to 0.2em, and limits line width to 58 characters. A warm-tinted reading overlay reduces visual stress. These exact values are drawn from published accessibility research.

Benchmarking evidence: Only Claude (U3 = 5/5) offers a dyslexia-friendly font. ChatGPT, Gemini, DeepSeek, and Kimi all scored 2/5 on font customization. ClearMind goes beyond font alone — it adjusts ALL display parameters simultaneously based on research-derived values.

Literature evidence:
- Rello & Baeza-Yates (2017): 7-14% character spacing improvement; line height and font size matter.
- BDA Style Guide (2014) [19]: recommends increased spacing and sans-serif or dyslexia fonts.
- BDA Typeface Review (2018) [20]: OpenDyslexic reduces b/d/p/q confusion.
- Wilkins (n.d.) [15]: visual stress / Meares-Irlen — tinted overlays reduce shimmer/blur.
- Miniukovich et al. (2019) [18]: minimum 16px font for web readability.

Code location: config.py DISPLAY_SETTINGS + index.html CSS mode overrides + local .otf font files

Benchmarking dimension addressed: U1 (Font Size), U2 (Line Spacing), U3 (Dyslexia Font)

Feature 7: User-Controlled Mode Selection (Agency Preservation)

ClearMind does NOT auto-detect disabilities. Users choose their mode via buttons in the header. They can switch modes at any time. This preserves user agency — a critical principle in disability research.

Benchmarking evidence: Custom instructions (U10) scored 1/5 for DeepSeek, meaning users must re-explain their needs every session. ClearMind's mode system eliminates this — select a mode once and the entire pipeline adapts.

Literature evidence:
- Tang et al. (2026): 'honor disabled ways of knowing' — do not impose assumptions.
- Glazko et al. (2025): users should control how AI adapts, not be forced into categories.
- Goodman et al. (2022): 'several rewrite choices rather than one imposed answer.'

Code location: index.html mode selector + config.py MODES

Benchmarking dimension addressed: U10 (Custom Instructions)

Feature 8: Output Sanitization (Thought-Process Leak & Repetition Prevention)

Free-tier AI models sometimes leak their internal chain-of-thought reasoning or enter repetition loops where the same phrase repeats dozens of times. ClearMind's output sanitizer strips thinking tags, detects repetition patterns, and truncates loops before the response reaches the user.

Benchmarking evidence: During testing, we observed instances where free-tier models returned internal reasoning tokens followed by repetitive fragments. No existing chatbot interface sanitizes these outputs before displaying them to users. For neurodivergent users, unexpected garbage output is especially disorienting.

Code location: gemini_client.py — _sanitize_output() function

Benchmarking dimension addressed: D9 (Error Recovery) — graceful handling of model failures

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Feature 9: Micro-Step Actionability (Task Initiation Support)

ClearMind's system prompts explicitly enforce micro-step instructions. When a user says 'I need to do X but I can't start,' the AI must give the FIRST PHYSICAL ACTION — not an abstract plan. Example: instead of 'Organize your workspace,' the AI says 'Pick up the closest item on your desk and put it where it belongs.'

Benchmarking evidence: Actionability (D8) was critically low for ChatGPT (2.0/5) and mediocre for most tools (3.0-3.5). Only Kimi scored 4.0. ClearMind's prompt rules explicitly ban abstract advice and require concrete micro-steps.

Literature evidence:
- Barkley (2012, 2015) [6, 7]: task initiation paralysis is one of ADHD's most debilitating symptoms. The executive function system cannot generate activation energy to begin.
- Giri et al. (2026) P17: 'overwhelm turns into confirmation of deeply held beliefs' — vague advice worsens paralysis.

Code location: prompts.py — rules 6, 11 (ADHD); rules 5, 13 (combined)

Benchmarking dimension addressed: D8 (Actionability)

Feature 10: Metacognitive Awareness (Alternative Format Suggestions)

When users indicate that reading is difficult or they are struggling with the format, ClearMind's prompts instruct the AI to offer alternatives: a shorter version, a list, an analogy, or a step-by-step walkthrough. It does not just repeat the same information in the same format.

Benchmarking evidence: Metacognitive awareness (D10) had a ceiling — no tool scored above 4.33. When users said 'reading about this is really hard for me,' most tools just gave a text explanation anyway, ignoring that reading IS the problem.

Literature evidence:
- Shaywitz & Shaywitz (2005) [13]: dyslexic users process spoken/visual information better than written text.
- Scoping review of AI for neurodivergent students (2025) [31]: AI tools should offer multiple modalities.

Code location: prompts.py — rules 10-11 (dyslexia); rules 12, 14 (ADHD/combined)

Benchmarking dimension addressed: D10 (Metacognitive Awareness)

Part 3: Code Changes Log

The following changes were made to the ClearMind codebase based on the benchmarking study findings:

File	Change	Benchmarking Gap Addressed
refocus.py	Rewrote drift detection messages from clinical/ passive-aggressive tone to warm, supportive language with 3 random variants. Added citation to Migliore et al. (2025) on rejection sensitive dysphoria.	D9 (Error Recovery) RSD awareness [10, 11]
gemini_client.py	Added _sanitize_output() function: - Strips <think>/<thinking> tags - Detects and truncates repetition loops - Catches short-phrase repetition within single lines - Wired into generate() pipeline	D9 (Error Recovery) Model output quality
prompts.py	Major prompt overhaul: - Added micro-step actionability rules - Added 120/150 word caps - Added yes/no-first rules - Added metacognitive awareness rules - Added explicit no-chain-of-thought rule - Updated docstring with benchmarking citations	D2 (Conciseness) D6 (Brevity) D8 (Actionability) D10 (Metacognition)
index.html	Added collapsible response sections: - CSS for toggle button + collapsible container - JS in addMessageHTML() wraps chunks 3+ in expandable container - Smooth animation (max-height transition) - 'Show N more sections' / 'Hide' toggle	U7 (Collapsible Responses) 1/5 for ALL 5 chatbots

Part 4: Complete Bibliography (46 Sources)

Benchmarking Study Sources (31):

[1] Song, P. et al. (2021). Prevalence of adult ADHD: A global systematic review. Journal of Global Health, 11, 04009.

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[19] British Dyslexia Association (2014). BDA Style Guide.

[20] British Dyslexia Association (2018). Typeface review findings.

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[28] LaMPost: AI-assisted email for dyslexia (2022). arXiv.

[29] LLMs for Web Accessibility: Systematic Review (2025). arXiv.

[30] LLMs to generate Easy to Read content (2024). Frontiers in Computer Science.

[31] Scoping review of generative AI for neurodivergent students (2025). Computers and Education Open.

System Design Literature (15 additional):

- Glazko, A. et al. (2025). Accessibility Barriers in AI Chatbots for Disabled Power Users.
- Silverman, A. et al. (2025). Gap between technology development and disability user experience.
- Tang, J. et al. (2026). Honor disabled ways of knowing — participatory AI design.
- W3C COGA (2021). Cognitive Accessibility Guidelines — familiar words, short sentences.
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- Malhotra, H. (2026). AI Accessibility for Neurodivergent Users — Benchmarking Study.

- CDC Plain Language Guidelines — Grade 6-8 for public health communication.
- WCAG 2.1 AA — Web Content Accessibility Guidelines.
- OpenDyslexic font project — open-source typeface for dyslexia.
- Flask web framework documentation.
- OpenRouter API documentation — free-tier model access.
- pysbd — Python sentence boundary disambiguation library.

This report documents the complete research and engineering basis for ClearMind. Every feature is grounded in measured gaps from our benchmarking study and published accessibility research. Effectiveness evaluation with diagnosed users is identified as future work.